

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

1. Define a piecewise function $f(x)$, defined as follows:

$$f(x) = \begin{cases} (+x)^{1/3} & \text{if } x > 0, \\ 1 & \text{if } x = 0, \\ (-x)^{1/3} & \text{if } x < 0. \end{cases}$$

Make a plot of $f(x)$ for $x \in [-2, 2]$ to show your definition of f is correct.

2. Write an efficient recursive function to compute

$$p_0(x) = 1, \quad p_1(x) = x - 1, \quad \text{for } n > 1 : p_n(x) = (2x + 1)p_{n-1}(x) + (x - 1)p_{n-2}(x).$$

Make sure your function can compute $p_{100}(x)$. Do **NOT** use `cached_function`.